

Message capacity and claim wording set the transition points of collective truth-finding in language-model networks

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Abstract

Whether human or large language model (LLM), an agent in discussion processes only a few of the others' contributions, bounded by cognition, context, or cost. LLM collectives can settle on a wrong consensus even from a correct majority; we ask how far that bound alone fixes their fate, as crowding parameter α : each agent accepts an input with probability $e^{-\alpha r}$ after r acceptances. Over 31,824 randomized queries, we found that an 8-billion-parameter model's judgment of a claim effectively reduces to a logistic function of a weighted sum of its inbox, equivalent to a stochastic binary neuron with divisively normalized weights. From these coefficients and degree statistics alone, the wrong-consensus basin should vanish at $\alpha^* = 0.435$ (6.4 of 31 sources). In 1,414 episodes with assigned opinions the prediction failed: correct-side outcomes stayed below 50% of episodes from every start, 28–45% from 75%-correct majorities. The failure traces to the field, the tilt each claim's wording induces: the experimental claims' fields lay below the calibration mean; with each claim's own field the same weights reproduce the outcomes. Reversing the wording showed the tilt follows a claim's assertion, not its truth. On a second 8B model the pipeline predicts claim-dependent bistability; basin boundaries appeared where computed, and eight-claim calibration matched in 15 of 16 conditions. At 70B the assertion bias is not detected. A collective's fate is thus largely set by two single-agent measurements: the tilt a claim's wording induces, and the message capacity that sets the transition point.

Significance statement

Language-model agents are beginning to operate in groups that debate and settle on answers, and such groups can talk themselves into the wrong one. This work shows that a group's fate is largely set by two quantities measurable on a single agent. An agent's judgment reduces to the input-output rule of a stochastic binary neuron with divisively normalized inputs, so the group's transition point follows, through a reduced map, from how many sources each member reads; the second quantity is the tilt a claim's wording induces on each reading. Across three models, collectives followed the assertion rather than the truth in one and crossed predicted transition points in another. AI collectives can be assessed, and designed, before they run.

1 Introduction

Groups of LLM agents that exchange messages and update binary opinions are now studied as statistical-mechanical systems. [El et al. \(2026a\)](#) showed that a kinetic Ising model fitted to one-step transitions describes more than 10,000 such communities ($N = 32$, eight rounds), predicts individual trajectories, and generalizes to unseen communication graphs. In that framework, as in most multi-agent opinion studies, the interaction graph is supplied by the

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experimenter; the question of where the graph comes from, and what its statistics do to the collective outcome, is listed by those authors as future work.

A separate line of work on synaptic resource allocation provides a one-parameter answer to the first half of that question. If a receiving unit accepts a further input with probability $e^{-\alpha r}$ once r inputs are already accepted, the number of accepted inputs grows only logarithmically in the candidate pool: the mean in-degree is $(\log N)/\alpha$ with bounded variance, the full in-degree distribution $P_{\alpha,N}(k)$ obeys a closed recursion, and the distribution is invariant under reordering of the candidates (Fukushima, 2026). The parameter α — the crowding parameter — compresses everything about the acceptance process that matters for degree statistics. Throughout, α describes the crowding of a finite intake budget, the message capacity of an agent, at the level of whole messages; it is unrelated to the attention mechanism inside transformer architectures, and we avoid the word “attention” for this reason.

The bound itself is not particular to machines. Human deliberation operates under cognitive limits of the same shape: working memory holds only a few items at once (Cowan, 2001), and divided attention caps how many contributions a participant actually processes (Hodas and Lerman, 2012). A theory that turns an intake bound into a collective outcome is, at least potentially, a statement about committees and juries as well as about machine collectives. In human groups, however, neither the bound nor the network it induces can be measured or controlled; in LLM collectives, both can.

These two ingredients suggest a quantitative question. Consider a population in which a fraction x_0 of agents initially holds the correct answer to a binary question. After several rounds of exchange the population typically reaches a correct or a wrong consensus; the probability of the former after a fixed number of rounds, $q(x_0)$, is the finite-horizon analogue of the committor of the collective dynamics (Methods). Mean-field theory on the generated networks predicts that the basin of the wrong consensus disappears entirely above a critical crowding level α^* : for $\alpha > \alpha^*$, even small correct minorities propagate. Our question is to what extent α^* can be computed in advance — from the degree statistics that α induces and from response coefficients measured on single agents in isolation — with no quantity fitted to collective runs. The nearest precedents come from LLM naming games, where emergent conventions and collective bias are established phenomena (Flint Ashery et al., 2025; Flint et al., 2026): a mean-field consensus condition has been assembled from single-token acceptance rates measured on isolated models (De Nobili et al., 2026), and a mean-field account of population-size-dependent collective bias from logit-extracted single-agent policies (Flint et al., 2026). These constructions share our logic, but they concern coordination on a convention (no correct side, no committor) under fully mixed or imposed interaction structures. No previous study has, to our knowledge, generated the interaction network of an LLM collective from a crowding process, or predicted the basin boundary of a truth-finding collective — the critical initial fraction of correct agents — from single-agent measurements completed before the collective ran.

Two families of confound make this comparison easy to get wrong, and both have shaped our design. First, in observational data the initial fraction x_0 is not an experimental dial: we show in Section 3 that in the largest public dataset x_0 is driven by question difficulty, which also drives the outcome, so the apparent committor is a difficulty curve and the transition point (the tipping point of the social-dynamics literature) is unidentified. Our collective experiment therefore assigns initial opinions exogenously. Second, changing the network changes the composition of each agent’s prompt, and LLMs respond to prompt composition regardless of any network; effects of this kind reproduce in a single agent with a single query. We therefore measure the single-agent response function first (Section 4), predict the collective outcome from it, and treat any residual — quantified by replaying every realized prompt through the measured response function — as the collective contribution proper.

The paper proceeds in three steps: a theory of α^* from a reduced map (Section 2), the

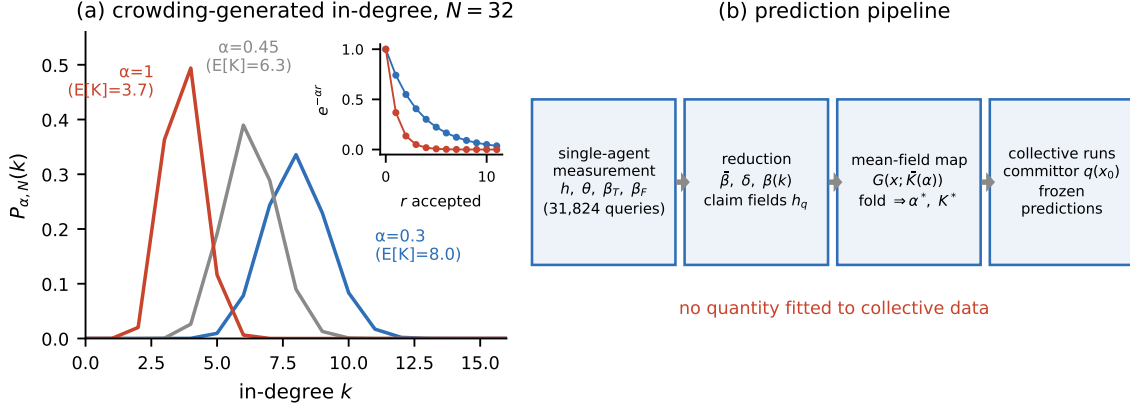


Figure 1: **Crowding-generated networks and the prediction pipeline.** (a) In-degree distribution $P_{\alpha,N}(k)$ generated by the acceptance process $e^{-\alpha r}$ (inset) at three crowding levels, $N = 32$. (b) Every quantity used to predict the collective outcome is measured on single agents or computed from the reduced map; nothing is fitted to collective data.

measurements that reduce the experimental agent to a stochastic binary neuron and give the theory its empirical coefficients (Sections 3–4), and a collective experiment run against the pre-registered prediction (Section 5). The pre-registered prediction fails, and the failure is traced to two measurable causes rather than to the collective step: the tilt each claim’s wording induces (its field), which did not carry over from the calibration claims to the experimental ones, and the eight-round horizon, which is short against the relaxation of the mean-field map near its fold. Two further campaigns then identify what dominated: an assertion-polarity bias specific to the model class, established by a pre-registered discrimination experiment and not shared by a second model — and in that second model the same pipeline predicts claim-dependent bistability whose basin boundaries are observed at the computed positions. To what extent a collective’s truth-finding is decided by individual input crowding is, on this evidence, a question with a computable answer — model by model, claim by claim.

2 Theory: a critical crowding level from a reduced map

2.1 Network generation

The generation process and its degree statistics are those of Fukushima (2026): each receiving agent processes the other $N - 1$ agents in arbitrary order and, having already accepted r sources, accepts the next with probability $e^{-\alpha r}$ (the first is always accepted); the in-degree distribution $P_{\alpha,N}(k)$ follows the recursion $P_{m+1}(r) = P_m(r)(1 - e^{-\alpha r}) + P_m(r-1)e^{-\alpha(r-1)}$, with mean $E[K] = (\log N)/\alpha + O(1)$. New here is only the closure: the deterministic continuum limit gives the finite- N form

$$\bar{K}(\alpha) = \frac{1}{\alpha} \ln(1 + \alpha(N - 1)), \quad (1)$$

which underestimates the exact mean by 3% ($\alpha = 0.3$) to 11% ($\alpha = 2$) at $N = 32$; where precision matters we invert the exact recursion instead. Eq. (1) is a large- N , moderate- α form: as $\alpha \rightarrow \infty$ it tends to zero, whereas the exact mean tends to one because the first candidate is always accepted. Degree statistics are invariant under candidate reordering, so manipulations of *who* is read change nothing in $P(k)$ — the property that lets order-based conditions be compared at exactly matched communication budgets.

2.2 Single-agent response and its reduction

Measurements in Section 4 show that an agent receiving l correct-side and $k - l$ wrong-side messages adopts the correct side with probability $\sigma(c_0 + \theta \mathbf{1}[\text{own state correct}] + \beta_T l + \beta_F(k - l))$, where σ is the logistic function (with the per-message weights attenuated by inbox size as $\beta g(k)$; Section 4); the measured unit is equivalent to a stochastic binary neuron. Three combinations organize everything that follows: the symmetry-breaking field $h = c_0 + \theta/2$, the tilt that the claim's wording induces on each reading, measured per claim (the persistence indicator decomposes into a constant field $\theta/2$ plus a symmetric term), the per-input swing $\bar{\beta} = (\beta_T - \beta_F)/2$, and the per-input drift $\delta = (\beta_T + \beta_F)/2$. The population dynamics under heterogeneous mean field (HMF) — each of an agent's k inputs correct independently with probability x — is the one-dimensional map

$$F(x) = \sum_k P_{\alpha, N}(k) \sum_{l=0}^k \binom{k}{l} x^l (1-x)^{k-l} [x \sigma(u_1) + (1-x) \sigma(u_0)], \quad (2)$$

with $u_s = c_0 + \theta s + \beta_T l + \beta_F(k - l)$; the interior unstable fixed point of F separates the two consensus basins. The construction is the mean-field analysis of the generating model (Fukushima, 2026) with one substitution: the hard-threshold unit assumed there is replaced by the measured logistic unit, of which the threshold unit is the infinite-gain limit; the outcome observable $q(x_0)$ is likewise inherited from that analysis, which computed the committor proper by an absorbing Markov-chain closure; here it is read after eight rounds (Methods). No threshold nonlinearity is assumed anywhere below — bistability is a property of the fixed-point structure of the S-shaped map, an unstable interior fixed point separating two stable ones — and the substitution removes the structural discontinuity that hard-threshold updates produce at truth-weight 1 (equal per-message weight for correct-side and wrong-side sources; Section 3). Exact numerics iterate Eq. (2) directly; closing the degree at $\bar{K}(\alpha)$ and smoothing the binomial with the standard Gaussian–logistic approximation yields the reduced map

$$G(x; K) = x \sigma(z_+) + (1-x) \sigma(z_0), \quad z_s = \frac{(h \pm \theta/2) + K\delta + 2\bar{\beta}K(x - \frac{1}{2})}{\sqrt{1 + \frac{\pi}{8} 4\bar{\beta}^2 K x(1-x)}}. \quad (3)$$

A degree-dependent coupling $g(k)$, for example the divisive normalization $\beta/(1 + \gamma k)$, one of the two attenuation forms our data support, enters by multiplying $\bar{\beta}$ and δ , leaving the form unchanged.

2.3 The critical capacity

α^* is the capacity at which the interior unstable fixed point of G undergoes a fold ($G = x$ and $G' = 1$ simultaneously). We keep the term critical crowding level for α^* for continuity with the generating model, but the transition is a saddle-node bifurcation of the mean-field map, corresponding to the disappearance of a metastable basin (a spinodal); the Discussion returns to the distinction. Operationally: the fold defines a threshold degree K^* , and α^* solves $\bar{K}(\alpha) = K^*$. The mechanism is visible in the linear gain at the symmetric point,

$$\lambda(K) = [\sigma(z_+) - \sigma(z_0)] + \frac{2\bar{\beta}K \bar{\sigma}'}{\sqrt{1 + \frac{\pi}{8} \bar{\beta}^2 K}}, \quad (4)$$

a persistence gain (first term, carried by θ) plus a social-amplification gain that, for constant coupling, grows only as $\sqrt{K}$ because input fluctuations smooth the response; here $\bar{\sigma}' = [\sigma'(z_+) + \sigma'(z_0)]/2$, with every term evaluated at $x = 1/2$. In the symmetric case ($h = 0$ and $\delta = 0$) bistability requires $\lambda > 1$. With a nonzero field the gain at the symmetric

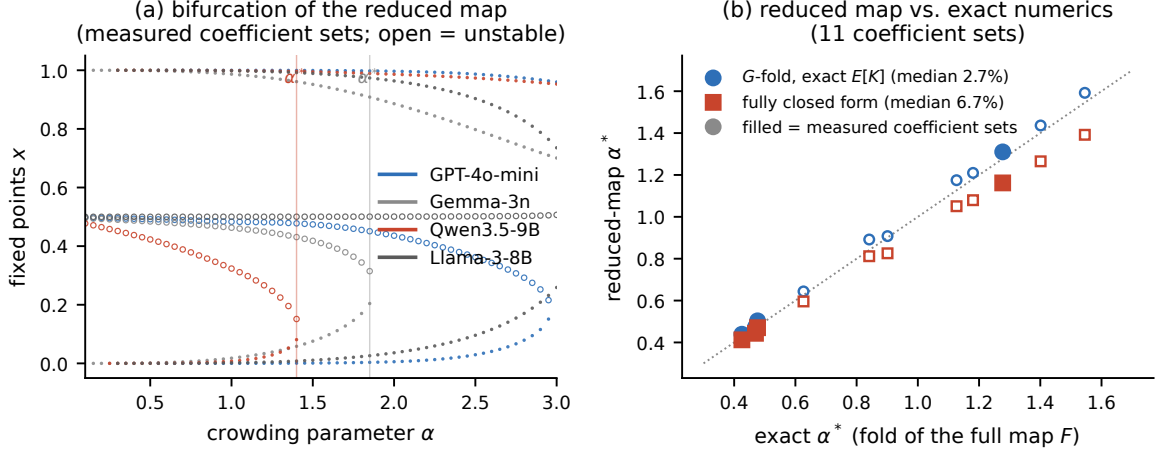


Figure 2: **Critical crowding level from the reduced map.** (a) Fixed points of the reduced map G as functions of α for the four measured coefficient sets of El et al. (2026a); the interior unstable branch (open symbols) vanishes at α^* . (b) Validation of the two closure levels against the fold of the full map F over 11 coefficient sets (filled symbols: coefficient sets measured in this study).

point locates the mechanism but is neither necessary nor sufficient for bistability, and it misplaces α^* by 30–90% at realistic fields, so all quantitative predictions below come from the fold of G . One further structural fact falls out of the reduction: with the measured $\delta < 0$ the effective field $h + K\delta g(K)$ is screened as degree grows (with constant coupling it crosses zero near $K \approx 10$ for the coefficients measured in Section 4 under the collective experiment’s prompt, variant A), which is why bistability survives at low α despite $h > 0$. Under divisive coupling this screening saturates: $K\delta g(K) \rightarrow \delta/\gamma$, and over the mean degrees realized here ($K \approx 3$ –10) the factor $Kg(K)$ varies only between 0.82 and 1.0, so the effective field, and with it the threshold position, depends only weakly on α . The weak α -dependence of the qwen3:8b thresholds (Section 5.4) is consistent with this; the zero crossing itself is not tested separately.

Against exact numerics (the full $P(k)$ sum, fold located by bisection in α) the closure is accurate to a median 2.7% over a 3-decade parameter grid and both measured coefficient sets when the exact $E[K]$ is inverted, and 6.7% in the fully closed form; the worst cases are 6.0% (exact $E[K]$; the synthetic set $h = 0.7$, $\theta = 0.6$, $\bar{\beta} = 1$) and 9.9% (closed form; $h = 0.3$, $\theta = 2$, $\bar{\beta} = 1$) (Fig. 2; SI). For the measured variant-A coefficients the prediction is $\alpha^* = 0.435$, $K^* = 6.4$, as registered (the fold conditions solved to full precision give 0.439 and $K^* = 6.36$, inside the registered tolerance): the wrong-consensus basin is predicted to disappear when agents accept, on average, fewer than 6.4 of their 31 possible sources. The critical crowding level of the collective is, to this accuracy, computable from single-agent coefficients through the reduced map.

3 Why public data cannot test the prediction

The natural first test requires no new data: El et al. (2026a) released full logs of 6,400 objective-question communities (four models, 40 questions, ten graphs per question, four episodes) (El et al., 2026b). We computed each community’s correct-side fraction trajectory $x(t)$ and asked whether the apparent committor $q(x_0)$ identifies a transition point.

It does not, for a reason that constrains observational designs in which the initial fraction is not assigned. Across the four models, between 50% and 99.5% of the variance of x_0 lies

between questions; within questions the x_0 -outcome correlation drops to 0.04–0.27. Communities at $x_0 \approx 0.40$ on different questions run to opposite consensuses with near certainty, while communities at $x_0 = 0.15$ on an easy question converge correctly in 32 of 32 episodes. Question difficulty drives both the initial fraction and the outcome; the pooled $q(x_0)$ curve is a difficulty curve, and its 0.5-crossing is not a property of the collective dynamics. A question-clustered bootstrap widens the apparent confidence intervals on the fitted crossing by a factor of 4–10, after which no pairwise difference among models or graph families is resolved. Two further obstacles are structural: the released graphs have nearly degenerate degree distributions (random family mass concentrated on $k = 2$ –4; lattices), so the degree-statistics content of any mean-field prediction is untestable; and under the hard-threshold update rule assumed in earlier work the predicted crossing jumps discontinuously at truth-weight 1, reducing the comparison to a single bit. These are limitations of the identification strategy, not of the dataset’s quality — the same logs support the message-level measurement below, which is what makes the rest of this paper possible.

The logs do yield the microscopic coefficients. Regressing each agent’s next-round correctness on the composition of its inbox (43,500 observations per model, question-clustered errors), the truth asymmetry w_T — the relative per-message weight of correct-side sources, previously estimated at 1.16–1.71 from Ising couplings — is indistinguishable from 1 in all four models (e.g. GPT-4o-mini: 1.055, 95% CI [0.996, 1.114]), while three other coefficients are robustly nonzero: a persistence term θ , an individual field c_0 , and an attenuation $\lambda_{\text{lab}} = 0.08$ –0.56 of messages arriving through “disagree”-labeled edges. The asymmetry that breaks the symmetric degeneracy of the mean-field map therefore resides in the field $h = c_0 + \theta/2$, not in message weighting — which is what Section 2 assumes. Public data fix the form of the theory; they cannot test its collective prediction.

4 Measuring the response function of the experimental agent

The collective experiment uses llama3.1:8b (4-bit Q4_K_M quantization, temperature 0.7, snap-judgement prompt matching El et al.’s protocol). We measured its response function directly: 31,824 single-shot queries in which the inbox composition (k messages, l of them correct-side), the A/B label assignment, the message order, and an optional explicit own-state line were randomized by design. Claims came from CLIMATE-FEVER (Diggelmann et al., 2020); a two-stage calibration under the bare judgment prompt selected 17 claims with per-claim field $|h_q| < 0.4$. Under the full experimental scaffold (persona and inbox lines) the fields did not transfer: the per-claim intercepts re-expanded to a range of 3.5 logits with mean 0.94, which is why the claims of the collective experiment were calibrated again under the full scaffold (Methods), and why the field, not the message weights, turns out to be the quantity that fails to carry across claim sets (Section 5.2). Sixteen of the 17 selected claims are REFUTES (claims labeled false in CLIMATE-FEVER); consequently all results are stated in correct-side/wrong-side terms, and the truth-versus-polarity distinction is explicitly not identified by this claim set: a diagnostic regression shows TRUE-side messages are more persuasive on average (ratio 1.44 [1.25, 1.65]), which accounts for the apparent $w_T = 0.74$ –0.89 < 1.

Three measured facts carry the predictions (Fig. 3; per- k coefficients and interval estimates in SI Fig. S1). First, the coefficients: without an explicit own-state line (variant A, the protocol of the collective experiment) $h = 0.725$ [0.516, 0.915]; with it (variant B) $\theta = 1.909$ [1.723, 2.106], consistent with the persistence measured in observational data being a persona-level carryover rather than an in-prompt self-input, since adding one line raises θ from absent to ≈ 1.9 . Fitted jointly with claim fixed effects anchored by the empty-inbox cells, the per-message coefficients fall from $\beta_T = 2.11$ [1.58, 2.96] and $\beta_F = -1.53$ [−2.03, −1.12] at $k = 1$

to 0.23 [0.18, 0.27] and -0.39 $[-0.48, -0.33]$ at $k = 12$ (SI Fig. S1); separate fits at fixed k do not identify these coefficients (Methods). Second, the load attenuation: adding messages attenuates each message’s influence, and model comparison (cross-validated by claim) favors inbox-size-dependent forms — load decay $e^{-0.14k}$ and divisive $\beta/(1+0.9k)$ are indistinguishable — over the position-dependent form $e^{-\alpha_{\text{pos}}r}$ closest to the theory’s acceptance kernel. The theory’s kernel is thus implemented by the exogenous router, not by the model’s internal weighting, under this protocol, in which message order is randomized; a deployment that orders messages by recency or salience could reinstate an internal position dependence. This demotion was fixed in the design before any collective run, and the internal decay enters the theory only through $\beta(k)$. The reduction itself is quantified twice: by claim-clustered cross-validated log-likelihood (-0.483 nats per observation for the two retained forms, against -0.509 for a constant-weight model with the same claim fields and self term and -0.693 for chance), and by the calibration of the fitted σ on every replayed collective transition (expected calibration error 0.03; Section 5). The same reduction is visible cell by cell as calibration: with divisively normalized weights, the observed correct-side rate in each of the 78 design cells matches the model’s mean predicted probability for that cell (RMS deviation 0.054 on the logistic curve), whereas the constant-weight model misses by inbox size (0.118), large- k cells lying between the curve and 0.5 and small- k cells beyond it (Fig. 3). The comparison is in-sample; the cross-validated log-likelihoods above are the out-of-sample statement. Third, the predictions: pushing the measured coefficients through Section 2 gives, as folds of the full map F , $\alpha^* = 0.43$ – 0.48 for variant A *under every reduction of the k -dependence* (constant 0.43, or 0.435 by the reduced map as registered; divisive 0.48; jointly identified per- k coefficients 0.46), while for variant B the reductions disagree — $\alpha^* = 0.47$ (constant) versus 1.28 (divisive). The disagreement is an experimental resource: an arm run under variant B between $\alpha = 0.6$ and 1.3 discriminates the coupling rule regardless of outcome. The response function of a single agent, measured in isolation, fixes every number the collective experiment is asked to reproduce.

5 Collective experiment

5.1 Design (pre-registered before any collective run)

Networks are generated per episode by the acceptance process at capacity α ; initial opinions are assigned exogenously ($\lceil 32x_0 \rceil$ agents seeded correct-side); dynamics follow El et al.’s protocol — one broadcast message per agent per round conditioned on its current stance, Markovian inboxes, snap-judgement opinion resampling — for eight rounds. The main arm (variant A prompts) runs $\alpha \in \{0.20, 0.30, 0.38, 0.45, 0.55, 1.00\} \times x_0 \in \{4, \dots, 24\}/32$, adaptively concentrated where q crosses $\frac{1}{2}$ and, above the predicted α^* , at low x_0 where the collapse of the wrong basin is most visible. The discrimination arm (variant B prompts) runs $\alpha \in \{0.60, 0.90, 1.30\}$. A rewiring control (in-degree-preserving shuffle with rewiring fraction $\rho = 1$, i.e., all sources resampled, at $\alpha = 0.30$) checks that outcomes are reproducible when every receiver’s sources are redrawn at fixed in-degree, a test of reproducibility under the mean-field premise rather than of its sufficiency (Section 5.2); a yoked one-step analysis replays every realized inbox through the Section 4 response function, so that any collective deviation from iterated single-agent behavior is measured rather than assumed absent; and a surrogate rollout of the measured per- k rule on the same graphs and seeds provides the finite-size prediction band. Decision criteria were fixed in advance: the empirical α_{emp}^* is the capacity at which the wrong-consensus rate from $x_0 \leq 8/32$, after eight rounds, falls through 20%, and the discrimination arm’s verdict is read from which α levels retain a wrong basin. Outcomes are read after eight rounds: an episode ends in a correct-side consensus when $x_8 \geq 0.6$ and a wrong-side consensus when $x_8 \leq 0.4$. The threshold is operational; among

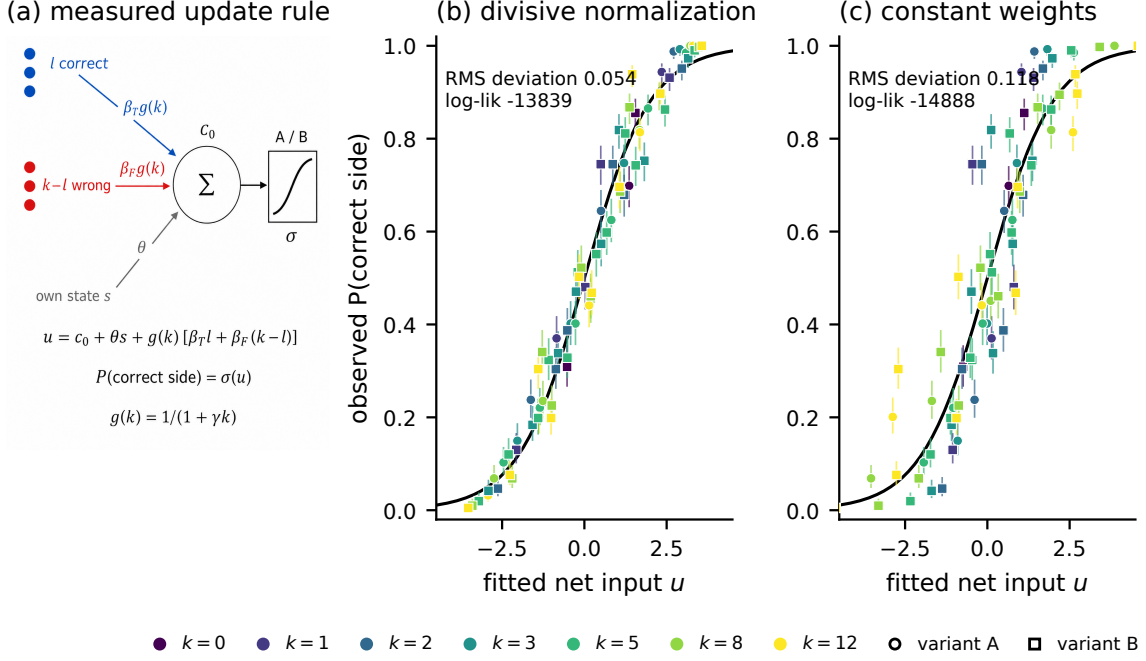


Figure 3: **Observed correct-side rate against fitted net input under divisive and constant message weights.** (a) The measured update rule: l correct-side and $k-l$ wrong-side messages enter with weights $\beta_T g(k)$ and $\beta_F g(k)$, the own-state indicator $s \in \{0, 1\}$ (variant B only) with weight θ , and the field c_0 ; the choice probability is $\sigma(u)$ with $g(k) = 1/(1 + \gamma k)$, $\gamma = 0.9$. (b) Observed correct-side rate in each of the 78 design cells of the 31,824 randomized queries (variant $\times k \times l \times s$; 408 queries per cell; Wilson 95% intervals) against u , the logit of the model’s mean predicted probability in the cell (divisive normalization, claim fixed effects), so that a model calibrated in that cell places it on the curve $\sigma(u)$. Color gives inbox size k ; circles variant A, squares variant B. (c) The same cells under constant weights ($g \equiv 1$): large- k cells lie between the curve and 0.5 and small- k cells beyond it. RMS deviation of cell rates from $\sigma(u)$: 0.054 in (b), 0.118 in (c).

decided episodes the final fraction lies beyond $|x_8 - 0.5| > 0.4$ in 63% of the main arm’s episodes (median wrong-side end state $x_8 = 0.03$) and in 97–99% of the qwen3:8b campaigns of Section 5.4, so the labels describe strongly polarized end states, and the sensitivity of every result to the threshold ($\delta \in \{0.05, 0.2\}$) is reported in the SI.

5.2 Results

The pre-registered prediction failed, and we state the failure before its diagnosis. Across all 1,414 episodes (zero parse failures), the correct-consensus probability never reached 0.5 in any (α, x_0) cell: the collectives converge on the wrong consensus from most initial conditions, and even from $x_0 = 24/32$ the correct-side outcome occurred in only 28–45% of episodes (wrong-side 33–73%, undecided 0–28%; Fig. 4a). The wrong-consensus rate from low x_0 falls monotonically with α — 0.98 at $\alpha = 0.20$ to 0.75 at $\alpha = 1.00$, the direction the theory predicts — but never crosses the pre-registered 20% threshold, so no empirical critical crowding level exists in the window: the prediction $\alpha^* = 0.435 \pm 0.03$ is rejected (claim-clustered bootstrap, 0 of 2,000 resamples crossing).

The diagnosis localizes the failure to the wording-induced field. Replaying every realized inbox through the Section 4 response function with each claim’s own field (per-message weights

pooled over the 17 calibration claims, field fitted per claim) reproduces the observed transitions without systematic error: Brier 0.10–0.14 against 0.17–0.25 for always predicting the base rate, expected calibration error 0.03, and a residual of -0.02 to $+0.01$ at contested compositions ($l/k \approx \frac{1}{2}$). The four experimental claims, chosen for near-neutral fields, carry fields of 0.16–0.55 under the scaffold; the calibration set’s fields average 0.94, and the blind rule inherited that average. Re-inserting the four fields into the unchanged theory predicts the observed phase: a single wrong-consensus attractor for three claims at every α in the window ($x^* = 0.02$ – 0.11) and, for the fourth, a wrong basin extending to $x_0 \approx 0.7$ up to $\alpha = 0.55$. Rolled out for eight rounds on the realized graphs and seedings, the same rule reproduces the wrong-consensus rate from low x_0 (0.93–1.00 predicted, 0.75–0.99 observed) and the final fractions to an RMS of 0.09 over the 36 cells; replacing the realized graphs by the mean-field independence assumption changes the eight-round prediction by a further 0.04 (RMS). We call this failure mode field transport: a field measured on one claim set does not carry over to another, whereas the message weights do. A second limit of the blind test is the horizon. The blind rule’s own eight-round surrogate on the realized graphs falls below 20% wrong consensus only at $\alpha = 0.63$, not at $\alpha^* = 0.435$ (at $\alpha = 0.45$ it still predicts wrong consensus in 67% of low- x_0 episodes; Fig. 4a, dashed), so the pre-registered criterion tested the fold through a horizon that displaces it by about 0.2 in α : the fold is a property of the infinite-time map, the experiment a statement about eight rounds, and the two are compared through the surrogate, not directly. The two causes separate on the counterfactual: with the blind field, the surrogate fails the 20% criterion at $\alpha \leq 0.55$ on its own (horizon), but at $\alpha = 1.0$ it predicts wrong consensus in 0.1% of low- x_0 episodes against 75% observed (field). The field is why the prediction fails across the window; the horizon is why the criterion could not have resolved the fold near α^* even with the right field.

The discrimination arm survives with a weaker verdict (Fig. 4b). Under the pooled coefficients the constant rule predicted no wrong basin at $\alpha \geq 0.6$ and the normalized rule predicted absorption up to $\alpha = 1.3$; wrong-consensus rates of 0.975–1.00 at all three capacities matched the normalized prediction, which was the pre-registered criterion. With claim-resolved fields, however, both coupling rules predict wrong-side attractors at these capacities, and the arm separates them less sharply: the normalized rule places the low- x_0 final fractions at 0.10–0.20 against 0.06–0.10 observed, the constant rule at 0.25–0.39. The data favor the normalized rule but do not reject the constant one, and one claim (1569) converged wrong-side where both rules predicted correct-side convergence. Degree-preserving rewiring ($\rho = 1$ at $\alpha = 0.30$) changed the outcome probability by at most 0.09 in any cell and by 0.00 in four of six; because the generated network is itself uniform given its degrees, this control tests reproducibility under redrawn sources rather than the sufficiency of $P(k)$. One residual remains: the collectives end slightly more correct than the iterated single-agent rule predicts (final fractions 0.05–0.17 above the surrogate in 11 of the 12 cells at $x_0 \geq 5/8$). The collective experiment thus rejects the blind prediction, verifies each link of the reduction chain except the transport of the field across claims, and identifies the claim’s field as the quantity that decided the outcome in the main arm.

5.3 What the field is: a pre-registered discrimination experiment

The dominant field admits three interpretations, which we separated with predictions registered before data collection: an assertion-polarity bias (collectives drift toward affirming the claim, whatever its content), a truth asymmetry (wrong content itself is favored), and a parametric-prior pull (collectives drift toward the model’s stored world knowledge). Two designs separate them. First, collectives on three *true* claims (labeled SUPPORTS in CLIMATE-FEVER): polarity and prior predict correct-side convergence, truth asymmetry predicts wrong-side. Second, collectives on minimally negated rewrites of the four original

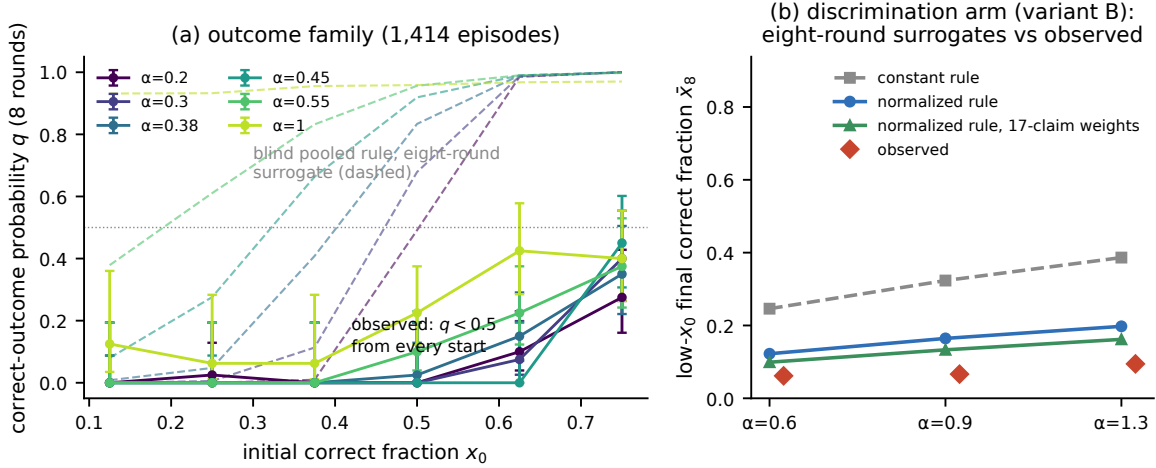


Figure 4: **Outcome probability after eight rounds against the surrogate of the pre-registered rule (arm A), and low- x_0 final fractions against surrogates of two coupling rules (arm B).** (a) Observed outcome family $q(x_0)$ after eight rounds (solid, Wilson 95% intervals) against the eight-round surrogate of the blind pooled rule on the same graphs and seedings (dashed): the blind prediction is rejected; the correct-side outcome probability stays below 0.5 from every initial condition. (b) Discrimination arm: observed low- x_0 final fractions against the eight-round surrogates of the normalized (divisive) and constant coupling rules with claim-resolved fields; the normalized rule lies closer, the constant rule is not rejected.

misinformation claims — each rewrite denies what its original affirms about the same proposition, so that the correct side switches from denying to affirming; three by inserting a negation and the fourth by replacing “a non-problem, or even a benefit” with “a problem, and not a benefit” (a contrary rather than a strict negation, with the quantifier “often” retained), equivalence checked by model judgment and by hand: polarity predicts the drift reverses with the wording; both alternatives predict it does not.

The verdict is polarity (Fig. 5). On all three true claims the collectives converged on the correct (affirming) side from every initial condition, including 12.5% correct starts (mean final fractions 0.67–1.00), rejecting truth asymmetry. On all four negated rewrites the collectives converged on the affirming side, which is now the correct side (correct-side fractions 0.98–1.00), exactly as the originals had converged on the affirming side when it was the wrong side (correct-side fractions 0.00–0.43): the assertion-referenced outcome is the same, the content-referenced outcome reverses with the wording, which a parametric-prior pull fixed by content cannot produce. The parametric-prior account is thus rejected in the form tested, a content-fixed prior read from the bare prompt (a quantity that, as Section 4 notes, does not itself carry over to the full scaffold), not in every conceivable form. One rewrite whose bare-prompt field pointed to the *denying* side ($p = 0.34$) still converged on the affirming side; this is the one pair in which field and polarity opposed each other, and the polarity channel overrode the field rather than merely riding it. Single-agent predictions again matched the collective outcomes claim by claim in all ten (claim, α) cells, two of them to within the resolution of the seeding grid (SI). A sender-side account, in which denying messages are simply weaker text, is excluded by two measurements: the tilt is present with an empty inbox, before any message is read, and the same 8B-generated message bank read by llama3.3:70b yields a per-message asymmetry of 1.03 [0.66, 1.64] (Section 5.4), so the asymmetry travels with the reader, not with the messages. For this model, the collectives follow the assertion, not the truth — which reframes the social reading of Section 5.2: the collectives lose to misinformation here because

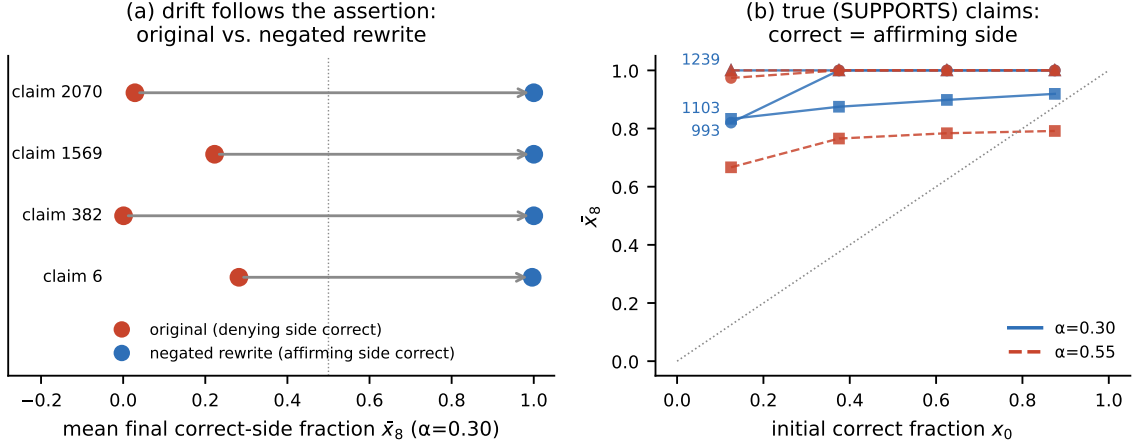


Figure 5: **Final correct-side fractions for original, negated, and true claims.** (a) Mean final correct-side fraction for the four original claims (denying side correct) and their minimally negated rewrites (affirming side correct) under identical conditions: the content-referenced outcome reverses with the wording. (b) True (SUPPORTS) claims converge on the correct = affirming side from all initial fractions.

misinformation is phrased as an affirmation, not because wrong content is favored. The same reading bears on reports that LLM opinion networks converge toward scientific ground truth regardless of initial stance (Chuang et al., 2024): where the true position is the affirming one, a polarity channel and a truth bias make identical predictions, and only designs that reverse the assertion direction — as here — separate them.

5.4 The regime is a model property

Running the identical pipeline — claim calibration and randomized single-shot measurement, then collectives, with fixed-point predictions computed from the single-shot fits alone — on a second 8-billion-parameter model (qwen3:8b) produced a qualitatively different outcome (Fig. 6). For all twelve qwen3:8b claims the single-agent data were collected before the corresponding collectives ran; the fixed points reported here are computed from those data with the final estimation procedure (Methods), so their agreement with the collectives is an out-of-sample test in which no collective outcome enters the fit, though not a prospective registration. The eight-claim campaign below adds the prospective element: its claim-selection rule was registered before its data existed. For this model the theory predicts, from the single-agent measurements alone, claim-dependent bistability: an interior unstable fixed point at 0.14–0.21 for one claim (falling with α and leaving the seeding grid at $\alpha = 1.0$) and at 0.70–0.71 for another. Both appeared. Collectives seeded at $x_0 = 0.125$ split between the two basins while every start at 0.375 or above converged correctly; for the second claim the correct-side outcome probability was at most 0.3 from every start at 0.625 or below (mean final fractions 0.00–0.42) and 0.9–1.0 from 0.875, placing the observed crossings at 0.69–0.75. These are the first observed interior thresholds in this study, at the computed positions. For the remaining two claims the pipeline predicts monostable correct-side convergence in every cell but one (the negated rewrite of claim 1569 at $\alpha = 0.30$, predicted bistable with threshold 0.37), and the collectives converged correctly in every cell but that one, where the observed crossing is 0.35 [0.28, 0.43]: all twelve cells match in class (one to within the resolution of the seeding grid), five of six predicted thresholds fall inside the observed intervals (the sixth misses by 0.02), and the prediction and observation intervals overlap in all six. This model is also not

assertion-following: one true claim converged to the *denying* side from all starts below 0.625, which no polarity account permits.

A calibration campaign then took the comparison beyond these four claims. Twenty further claims were calibrated and measured single-shot; the same pipeline classified each (claim, α) cell — monostable correct-side, monostable wrong-side, or bistable with a computed threshold — eight claims spanning the predicted regimes were selected by a pre-registered rule applied to the classifications computed at the time of the campaign (the predictions reported here, recomputed with the estimation procedure of Methods, leave the selected claims’ classes unchanged in 15 of 16 cells), and 256 further episodes were run ($\alpha \in \{0.30, 0.55\}$, eight initial fractions, zero parse failures). The predicted classification matched the observed one in 15 of 16 cells (Fig. 7). Across the eleven cells observed bistable, all eleven predicted thresholds fall inside the episode-bootstrap intervals, the mean absolute difference between predicted and observed thresholds is 0.04, and the two are rank-correlated at $r_s = 0.96$ (Spearman; $p = 0.004$ under permutation of claim labels, claims being the exchangeable unit). Prediction-side uncertainty, from resampling each claim’s single-shot queries, gives intervals of width 0.08–0.25 on the predicted thresholds that overlap the observed intervals in all eleven cells (Fig. 7, horizontal bars). The one mismatch is a claim at $\alpha = 0.30$ whose predicted threshold, 0.09, lies just above the lowest seeded fraction (2/32), from which its collectives nevertheless converged correctly; at $\alpha = 0.55$ the same claim’s predicted threshold (0.03) and observed crossing (0.01 [−0.09, 0.17]) agree, and it is bistable in only 62% of the resampled predictions, marking it as the boundary case. What was an existence proof on two claims is thus a calibration curve on eight: the single-agent pipeline predicts not only whether a threshold exists but, to the resolution of the seeding grid, where it sits.

The larger model completes the comparison. In llama3.3:70b, probed at the single-agent level with the same eight claims and message bank, the per-message assertion asymmetry is 1.03 [0.66, 1.64], against 1.44 [1.25, 1.65] for llama3.1:8b over the 17 calibration claims and 1.94 [1.39, 2.73] on these same eight (difference 0.90 [0.00, 1.72], claim-clustered), and the bare fields align with content for seven of the eight claims (with an empty inbox, $P(\text{TRUE}) = 0.000$ for the four false originals and 1.000 for three of the four true rewrites; the eighth, the negated polar-bear rewrite, is near neutral at $P(\text{TRUE}) = 0.42$): the polarity channel that governed the 8-billion-parameter collectives is not detected, and the larger model’s collective regime — not measured here, a stated limitation — would be field-dominated on most realistic claims. The larger model differs from llama3.1:8b in model generation as well as in scale, so the comparison bounds the channel in the larger model rather than isolating scale as its cause. Whether a collective has a transition point at all, where it sits, and what force drives consensus are, across the two models whose collectives were run, and for the polarity channel in the third, properties measurable on the individual agent before any collective is run.

6 Discussion

The intake budget is a design variable for collective reliability: through $P(k)$ it sets whether a wrong-consensus basin exists, and the critical budget K^* is computable from single-agent coefficients through the reduced map — on the precondition that the wording-induced field and any model-specific polarity channel are measured first, since these decide which phase diagram the budget is operating in. The evidence is three models run against predictions computed from single-agent measurements alone: assertion-dominated monostability in llama3.1:8b, where field and polarity override the budget and 75%-correct majorities are lost in most episodes; claim-dependent bistability with predicted thresholds in qwen3:8b (15 of 16 cells); no detectable polarity at 70B. A blind prediction, its pre-registered rejection, the replay-based localization of the failure, and the cross-model confirmation of the pipeline with claim-resolved

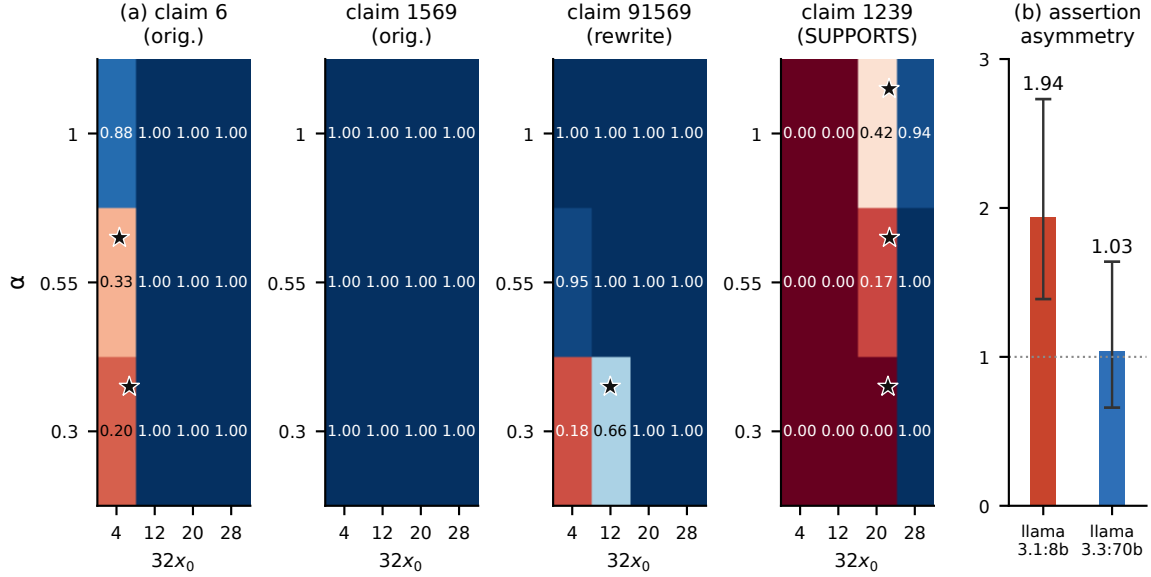


Figure 6: **qwen3:8b collective outcomes against predicted fixed points, and the per-message asymmetry at 70B.** (a) qwen3:8b collectives: mean final correct-side fraction per cell; stars mark the predicted unstable fixed points, which coincide with the observed basin boundaries for claims 6, 1239, and 91569 at $\alpha = 0.30$; claim 1569 is predicted and observed monostable correct-side. Cell values are mean final fractions; intermediate values are splits between the two basins, not intermediate end states (per-cell outcome counts in the SI). (b) The per-message assertion asymmetry on the same eight claims and message bank (claim-clustered 95% intervals) is not detected at 70B.

fields are reported together. Throughout, the α trend of the wrong-consensus attractor — observed and predicted — shows that the number of messages each agent was actually shown remained the operative degree even where the blind prediction failed: the failure mode was the transport of the field across claim sets, not the identity of the degree.

The blind test carries two lessons for prediction in this setting. The field must be measured on the claims that will be debated: message weights pooled over a calibration set transfer, the field does not. And the decision criterion must be computed for the horizon of the experiment: the fold of the mean-field map is an infinite-time statement, and near the fold the map relaxes slowly, so an eight-round experiment must be compared with an eight-round surrogate rather than with the fold itself. What the collectives verified should also be stated precisely. The observed transitions are thresholds in the initial fraction at fixed capacity, at the positions the pipeline computed. The fold in α itself, the vanishing of the wrong basin as capacity is reduced, was predicted for llama3.1:8b and not observed, because the claims' fields placed three of the four claims in the wrong-side monostable phase at every capacity tested and the fourth in a bistable phase whose threshold lay near $x_0 \approx 0.7$. The capacity dependence that was observed is the movement of the qwen3:8b thresholds with α (claim 6: 0.21 to 0.14 predicted, 0.19 to 0.17 observed, then below the seeding grid at $\alpha = 1.0$), which is the same fold approached from the side of the initial fraction. The title's claim is the first of these: capacity sets where the threshold sits, verified in qwen3:8b, and the wording sets which phase the collective is in; whether capacity can remove the threshold altogether, the fold in α , is the prediction that llama3.1:8b's fields kept out of reach.

What α corresponds to outside this experiment differs by system. Here α is an enforced reading budget: in this protocol the router, not the model, implements the acceptance kernel,

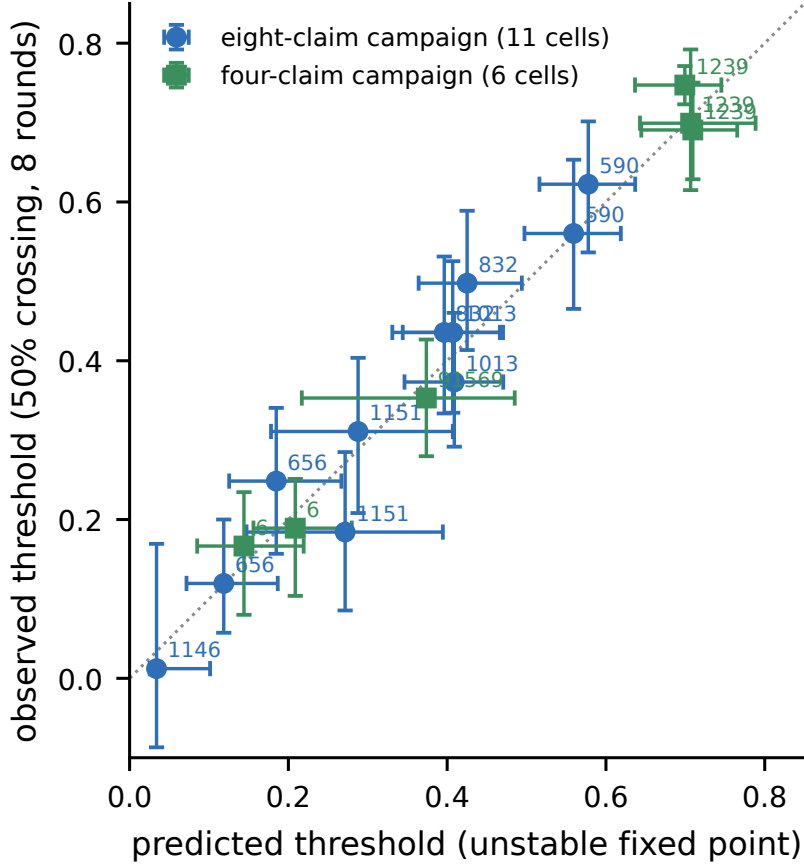


Figure 7: **Calibration of threshold predictions (qwen3:8b)**. Predicted unstable-fixed-point position (per-claim divisive response function) against the observed threshold (50% crossing of the eight-round outcome) for every (claim, α) cell observed bistable: circles, the eight-claim campaign (11 cells); squares, the four-claim campaign (6 cells). Vertical bars are episode-bootstrap 95% intervals of the observed threshold, horizontal bars are single-agent-bootstrap 95% intervals of the prediction, and the dotted line is identity. Cells predicted and observed monostable (4 of 16 and 6 of 12) and the one class mismatch (claim 1146 at $\alpha = 0.30$) are listed in the SI.

a division of labor the measurements themselves selected (Section 4). In deployed multi-agent systems its counterparts are the quantities that bound how many messages an agent reads — finite context windows, per-token costs, and the sampling policies of agent frameworks; in the naming-game literature the analogous bound is a finite interaction memory (Flint et al., 2026); in human information sharing, divided attention limits message visibility in the same direction (Hodas and Lerman, 2012). The model’s own capacity enters the theory separately, as the measured load attenuation $\beta(k)$: the intake budget sets who is read, the attenuation sets how much each reading counts.

Dialogue does not correct error in the field-dominated phase: majorities of 75% correct agents ended correct-side in fewer than half of the episodes at every capacity tested, and the discrimination experiment shows the drive is the assertion, not the content. Communication volume and capacity are no remedy where the per-reading tilt is adverse — a limiting condition on idealized-Bayesian accounts of collective knowledge (Taniguchi et al., 2025). The misinformation-vulnerability reading must be stated carefully: small llama collectives lose to misinformation because misinformation affirms; models without the polarity channel (qwen,

70B) fail or succeed by different routes. Directional conformity has likewise been reported to collapse majority-vote aggregation in LLM safety panels under peer pressure (Hu and Qu, 2026); the polarity channel measured here is a claim-level analogue that requires no peer opinions at all — it is present in bare single-shot calibration.

The origin of the polarity channel is probed and bounded rather than resolved. An ablated (refusal-direction-ablated, Arditi et al., 2024) derivative of the same instruct model retains the channel — per-message asymmetry 1.24 [1.05, 1.47] against 1.44 [1.25, 1.65] for the intact model, measured in the same batch with the same message bank — so removing refusal behavior does not remove the assertion bias, and any reduction is below this measurement’s resolution. Whether the channel exists before instruction tuning could not be measured: the base model does not follow the forced-choice format (parse rate below 5% even with a three-shot format prefix), which places pre-tuning behavior outside any forced-choice probe; a log-probability readout is the instrument for that question. The channel is adjacent to, but not identical with, sycophancy as usually defined — agreement with a user’s stated opinion, traced in part to the preference data used for RLHF (Sharma et al., 2023); here the agreement is with the claim’s own assertion, with no user opinion in the prompt. What can be said is negative and useful: the bias that decides these collectives’ fate is not carried mainly by the safety-refusal direction.

Two further single-agent observations are recorded without mechanistic claims: opinion flips concentrate in particular agents rather than being suppressed after a flip (a positive “anti-refractory” coefficient, +0.98), and the measured branching ratios (1.8–3.6) place these synchronous dynamics in a drift-dominated, far-from-critical regime — closing, for now, a self-organized-criticality reading of these collectives.

What is verified here is the reduction chain from single-agent response and degree statistics to the basin boundaries of *LLM* collectives at fixed capacity; the fold in capacity itself remains a theoretical prediction. The suggestion for human groups is nonetheless concrete. The cognitive limits cited in the Introduction give human participants a finite effective intake of the same shape, and the theory’s inputs (response tilt, load attenuation, intake bound) are all behavioral, none tied to a machine substrate. If deliberating groups sit at finite effective α , the theory would say which of them can recover the truth from a correct minority and which cannot. Statements about human collectives are implications, not measurements: locating a human group on the computed α^* curve would need, besides an estimate of a human α from intake-saturation data, the human counterparts of the response coefficients (the tilt, the per-message weights, and their load attenuation), none of which is measured here. A further limitation is the REFUTES-skew of the calibrated claim set, which restricts all claims to correct-side/wrong-side language; identifying genuine truth asymmetry needs a claim set balanced in polarity at matched difficulty, which this model’s one-sided accuracy on SUPPORTS claims may make impossible — with the mitigation that the α^* prediction never invoked truth asymmetry: the theory weights correct-side and wrong-side messages by their measured coefficients, and the measured inequality $|\beta_T| \neq |\beta_F|$ ($\delta < 0$) is, for this REFUTES-skewed set, the assertion asymmetry seen from the correct side, not a truth asymmetry.

Relative to El et al. (2026a), their graphs were given, ours are generated; their fits describe the collective data, ours are computed from single-agent data alone. The two studies probe orthogonal axes of one phase diagram for the same class of system — binary units with logistic (Glauber-type) updates. Their transition is thermal: topology fixed, noise swept, a critical point located by a susceptibility peak, on symmetric couplings that admit an energy function. Ours is topological: effective noise fixed (absorbed in the measured β), degree statistics swept through α , and the transition is a fold — the disappearance of a metastable basin, a spinodal rather than a critical point, which is why the collapse of the wrong basin and not a fluctuation peak is the correct observable. Our graphs are directed, so the dynamics are non-equilibrium and free-energy arguments do not transfer; the mean-field map does. Model-

dependent collective regimes have now been reported in the coordination setting as well — naming-game populations whose collective bias amplifies, induces, or reverses individual bias with model-specific size thresholds (Flint et al., 2026) — convergent with the model dependence measured here; and the structured interaction topologies those authors list as future work are what the crowding process supplies. For consensus-scaling observations (De Marzo et al., 2024), $(\log N)/\alpha$ supplies the analytic form for the reported size-limited consensus.

Part of the logistic form is guaranteed by the output mechanism: for a forced choice between two options, the probability of one option conditional on the answer being one of the two, $P(A \mid A \text{ or } B) = \sigma((z_A - z_B)/T)$, is exactly a logistic function of the logit difference at temperature T ; the response format asks for a single character and parse failures were zero in every collective campaign, so the measured choice probabilities are close to this conditional quantity. What the mechanism does not guarantee — the empirical content of the reduction — is that this logit difference is approximately additive in the inbox composition, one weight per correct-side and per wrong-side message (Fig. 3b). Additivity in log-odds is what an aggregator of conditionally independent evidence would produce; whether the transformer approximates such a computation, and why, is left open alongside the mechanism of the attenuation below.

The measured update rule (Fig. 3a) departs from a classical Ising spin in exactly one place: the per-message coupling is attenuated by the total inbox size, $\beta/(1 + \gamma k)$, the form known in sensory neuroscience as divisive normalization (Carandini and Heeger, 2012). As in that literature, the law is phenomenological — a behavioral input–output relation, not an identified mechanism. Three candidate mechanisms make separable predictions: softmax attention inside the transformer is itself a divisive operation, and predicts attenuation governed by token volume rather than message count; proportion coding — judging by the fraction l/k of supporting messages rather than their number, the saturation pattern classically reported for human majority influence — predicts dependence on l/k alone; and calibration compression of the output probabilities would mimic attenuation and is excluded or quantified by reading log-probabilities directly. A factorial manipulation of message length and proportional (l, k) scaling within the Section 4 protocol would separate the three; we leave this as a stated open question, since no result in this paper depends on which mechanism generates the measured form.

The practical rule this work supports is short: measure the tilt first, then set the budget. Both are single-agent measurements, and together they account for most of what the collectives did, with residuals of 0.05–0.17 in final fraction and one claim outside both coupling rules remaining. The residual is small but systematic (11 of 12 cells, same sign), so it measures a collective contribution beyond the iterated single-agent rule rather than noise. One candidate is the evolution of message content across rounds: the influence of correct-side messages grows by +0.026 per round in the one-step regression (SI), which could reflect either the messages or repeated exposure. Replaying single-shot queries with round-specific message banks built from the logged episodes would separate the two; the logs permit it.

Methods

Models and serving

All agents are served with ollama. The primary model is llama3.1:8b (Q4_K_M quantization, temperature 0.7 throughout, seed recorded for every call). The cross-model campaign uses qwen3:8b (tag 500a1f067a9f, thinking disabled via `think:false`) and llama3.3:70b (Q4_K_M; judgments only, message banks fixed to the 8B-generated banks). Single-shot campaigns ran against a user-space ollama instance with `OLLAMA_NUM_PARALLEL=16` (the system instance saturates at 0.62 generations/s; the parallel instance reaches 3.5 generations/s); the 70B

model does not load under 16-slot KV allocation and was served by the serial system instance. The polarity-origin probe (Discussion) used two further variants, judgments only: llama3.1:8b-text-q4_K_M (the base model) and an abliterated derivative of the instruct model (mannix/llama3.1-8b-abliterated, Q4_0), with a three-shot answer-format prefix applied identically to all variants, including the re-measured instruct control. Every call’s prompt hash, seed, raw response, and parse result are logged to JSONL.

Prompts

The judgment prompt (verbatim; variant A) is:

```
# Your Expertise
{persona}
# Messages
You received the following messages from fellow panelists:
- {message 1} ...
# Problem
You are one member of an expert panel judging the following claim.
Claim: "{claim}"
Options:
A) {TRUE or FALSE}
B) {FALSE or TRUE}
# Instruction
Give your direct answer on which option is correct.
Do NOT work through the problem, show any steps, compute, or
explain --- just commit to a snap judgement.
## Response Format
Respond with a single character, A or B, and nothing else.
```

Variant B inserts, before # Problem: “# Your Current Answer / Your current answer is that the claim is {TRUE|FALSE}.” The A/B $\leftrightarrow$ TRUE/FALSE assignment is randomized per call and counterbalanced within cells; message presentation order is randomized and recorded. The message-generation prompt asks the sender, given its persona, the claim, and its current stance, to “write a brief message to a fellow panelist stating that you believe the claim is {stance} and giving the single strongest reason from your reasoning,” with a two-sentence response format; stance is expressed in the message body (TRUE/FALSE), never as an option letter, because option labels are randomized per receiver. Personas are one-line expertise statements drawn from a fixed pool of 12. Prompt texts are extracted verbatim from the released scripts.

Claims and calibration

Claims are drawn from CLIMATE-FEVER (Diggelmann et al., 2020) (SUPPORTS and REFUTES only; the claim text is presented alone, without evidence). A two-stage calibration (24 calls per candidate, refined to 64 for candidates with $p_q \in [0.35, 0.65]$; A/B assignment counterbalanced) measured per-claim accuracy p_q and field $h_q = \text{logit}(p_q)$. The collective experiments use per-claim fields re-estimated under the full experimental scaffold (Section 4). The negated rewrites of the four Stage-2 claims (each rewrite denies what its original affirms about the same proposition; three by inserting a negation, one by replacing “a non-problem, or even a benefit” with “a problem, and not a benefit”, a contrary that avoids a double negative and retains the quantifier “often”) were checked for semantic opposition by model judgment (70–85% of 20 counterbalanced calls per pair) and approved by hand before any collective run. Full claim texts, rewrites, and calibration values are in the SI.

Networks, episodes, and controls

Each episode generates a fresh directed network: every receiver processes the other 31 agents in an independent uniformly random order, accepting the next candidate with probability $e^{-\alpha r}$ after r acceptances. Initial stances are assigned exogenously to a uniformly random subset of $\lceil 32x_0 \rceil$ agents. Each of eight rounds consists of a message phase (every agent with at least one out-edge generates one two-sentence message conditioned on its current stance, broadcast to all its receivers) and a judgment phase (each agent answers the snap-judgement prompt with its current-round inbox; single sample). The network is fixed within an episode (regenerated only between episodes), so a receiver reads the same senders in every round, and their states are correlated through the shared graph even though every message is regenerated each round from its sender’s current stance; the independence assumption of the mean field of Section 2 is therefore an approximation. Its cost is quantified in Section 5.2 by comparing the eight-round iteration of the mean-field map with surrogate rollouts of the same rule on the realized graphs (RMS difference 0.04 in the final fraction). A fidelity check re-ran ten episodes with five-sample majority voting; absorption was sharper (wrong-consensus rate 1.00 versus 0.875, mean $|x_8 - 0.5|$ 0.44 versus 0.38), so the single-sample simplification is not the source of wrong-side absorption. The rewiring control resamples each receiver’s sources uniformly while preserving its in-degree. Outcomes are classified at $\delta = 0.1$ (correct consensus: $x_8 \geq 0.6$; wrong: $x_8 \leq 0.4$), with $\delta \in \{0.05, 0.2\}$ and undecided-exclusion sensitivity analyses in the SI.

Cross-model calibration campaign

Twenty further CLIMATE-FEVER claims (ten SUPPORTS, ten REFUTES, disjoint from all earlier sets) were calibrated under the full scaffold and measured single-shot with qwen3:8b (24 calls per candidate, 64 in the boundary band; per-claim response-function fits from 6,720 further queries). The resulting fixed-point classifications and thresholds for $\alpha \in \{0.30, 0.55\}$ were computed at the time of the campaign, and eight claims were selected by a rule fixed in the pre-registered instructions (at least four predicted bistable with interior thresholds, the remainder monostable, filled from the bistable pool where a category was empty — no wrong-side-dominant claim existed among the twenty candidates). Collectives ran the full grid $\alpha \in \{0.30, 0.55\} \times x_0 \in \{2, 6, \dots, 30\}/32$ with two replicates (256 episodes, protocol identical to the main cross-model campaign). Observed thresholds are defined under Statistics.

Statistics

All confidence intervals on measured coefficients are claim-clustered pairs bootstrap percentiles (1,000–2,000 resamples for the pooled coefficients, 200 for the identified per- k coefficients and for the prediction-side intervals, 2,000 for the polarity ratios; claims are the clusters because episodes within a claim share its field). Attenuation-form comparison uses five-fold cross-validation with folds split by claim, alongside AIC; the form parameter (γ for divisive normalization, the rate for load decay) is profiled on the full data ($\gamma = 0.9$) and re-profiled inside each training fold, and a held-out claim’s fixed effect, which cannot be estimated from the training folds, is set to zero, so the cross-validated likelihood scores prediction for an unseen claim with an unknown field. The divisive α^* of Section 4 uses γ profiled separately per variant (0.53 for A, 1.32 for B); Fig. 3 uses the pooled $\gamma = 0.9$. Per- k coefficients are identified by a joint logistic fit over all inbox sizes with claim fixed effects anchored by the empty-inbox cells: at a single k the columns 1, l , and $k - l$ are collinear, so separate per- k fits identify only $c_0 + k\beta_F$ and $\beta_T - \beta_F$. Per-claim response functions for the collective claims use the reduced form $c_0 + g(k)[\delta k + \bar{\beta}(2l - k)]$ ($+\theta s$ for variant B), with a weak ridge penalty (10^{-3}) where k takes three values. For Fig. 3, one logistic fit per prompt variant with claim fixed effects and message weights $\beta_T g(k)$, $\beta_F g(k)$ (divisive, $\gamma = 0.9$ fixed, or $g \equiv 1$)

assigns each query a predicted probability; a design cell’s net input u is the logit of its mean predicted probability, and its observed correct-side rate is plotted against u with a Wilson 95% interval. Fixed points of the mean-field map are bracketed on an endpoint-inclusive grid of 4,001 points in x with bisection refinement and classified by $|F'(x)| < 1$; a phase is bistable when an unstable interior point separates two stable ones, and α^* is located by bisection in α on that classification, with the fold of G solved jointly from $G = x$ and $G_x = 1$. Prediction-side intervals resample each claim’s single-shot outcomes within that claim and within each design cell (200 resamples; a within-claim bootstrap, not the claim-clustered bootstrap used for pooled coefficients) and recompute the fixed points. Reanalysis statistics on the public dataset use question-clustered bootstrap and CR0 sandwich errors.

Outcome and threshold. An episode’s outcome is its correct-side fraction after eight rounds, x_8 ; $q(x_0)$ is the fraction of episodes with $x_8 \geq 0.6$, a finite-horizon quantity that we compare with the infinite-time fixed points only through surrogate rollouts. Observed thresholds are the 50% crossings of a logistic fit of the correct outcome on x_0 with a weak ridge penalty (0.002) so that separated cells have a finite estimate; the slope is not constrained, a fit with a non-positive slope would be discarded, and none occurred in the reported cells. Intervals are episode-bootstrap percentiles (500 replicates); replicates without a finite crossing in $(-1, 2)$ are dropped, which retained all replicates in 12 of the 17 cells observed bistable, 97–99% in three, and 89% and 65% in the two cells whose crossing lies nearest the lowest seeded fraction (claims 656 and 1146 at $\alpha = 0.55$). Rank correlations between predicted and observed thresholds are tested by permuting claim labels, claims being the exchangeable unit.

Pre-registration timeline

Throughout this paper, “pre-registered” means that a prediction or a decision rule, together with the criteria for judging it, was written to a file before the corresponding data were collected, and that the file’s timestamp and SHA-256 hash are recorded in the project repository, so that the order of prediction and data is documented; these are local records, not a third-party time stamp. No external registry (such as OSF) was used. Three items were pre-registered: (i) the $\alpha^* = 0.435 \pm 0.03$ prediction and the decision criteria of the main collective experiment, before any collective episode (2026-08-23); (ii) the three-hypothesis prediction table of the polarity discrimination experiment, before its data collection (2026-08-25); (iii) the claim-selection rule of the eight-claim calibration campaign, written into the campaign instructions before its data were collected (2026-08-27). The single-agent measurements preceded the collective experiments. The claim-resolved predictions of Sections 5.2–5.4 were computed after the collective experiments, with the estimation procedure described under Statistics, using only the single-agent data for parameter fitting; they were not prospectively registered. The one adaptive element (second-pass allocation of episodes) followed rules fixed in the pre-registered design.

Use of generative AI

The author used Anthropic’s Claude for code generation and debugging, execution of the pre-specified experiment pipelines, drafting and editing of text, and literature search. The research questions, theory, experimental designs, pre-registered predictions, and interpretations are the author’s; all AI-assisted output was reviewed and fact-checked, and every reported number is recomputed from the logged data by the scripts in the public repository. The language models studied here (llama3.1:8b, qwen3:8b, llama3.3:70b) are the subject of the research, not writing tools.

Data and code availability

All episode logs (stance histories, full message texts, graphs, seeds, prompt hashes), single-shot logs, analysis scripts (scripts 50–57 produce every identified fit, fixed point, surrogate rollout, and figure reported here), and notebooks are retained in the project repository, to be released publicly on GitHub (https://github.com/dockmfgit/AI_agent_crowding); the public dataset of El et al. (2026a) is not redistributed and is available from its original source (El et al., 2026b). Layout: `scripts/` (measurement, engines, analysis), `results/` (all tables backing the figures), `data/` (raw JSONL and episode JSON), `notes/` (working records including the pre-registration files).

References

- Andy Arditi, Oscar Obeso, Aaquib Syed, Daniel Paleka, Nina Panickssery, Wes Gurnee, and Neel Nanda. Refusal in language models is mediated by a single direction. arXiv:2406.11717, 2024.
- Matteo Carandini and David J. Heeger. Normalization as a canonical neural computation. *Nature Reviews Neuroscience*, 13:51–62, 2012. doi: 10.1038/nrn3136.
- Yun-Shiuan Chuang, Agam Goyal, Nikunj Harlalka, Siddharth Suresh, Robert Hawkins, Sijia Yang, Dhavan Shah, Junjie Hu, and Timothy T. Rogers. Simulating opinion dynamics with networks of LLM-based agents. In *Findings of NAACL 2024*, 2024.
- Nelson Cowan. The magical number 4 in short-term memory: A reconsideration of mental storage capacity. *Behavioral and Brain Sciences*, 24(1):87–114, 2001. doi: 10.1017/S0140525X01003922.
- Giordano De Marzo, Claudio Castellano, and David Garcia. AI agents can coordinate beyond human scale. arXiv:2409.02822, 2024.
- Cristiano De Nobili, Vijayasri Iyer, Alessandro Codello, and Raffaella Burioni. Microscopic dynamics of consensus formation in multi-agent LLM naming games. arXiv:2608.02178, 2026.
- Thomas Diggelmann, Jordan Boyd-Graber, Jannis Bulian, Massimiliano Ciaramita, and Markus Leippold. CLIMATE-FEVER: A dataset for verification of real-world climate claims. In *Tackling Climate Change with Machine Learning workshop at NeurIPS 2020*, 2020.
- Batu El, Jinhee Paeng, Fatih Dinc, Shiye Su, Mete Erdogan, Aneesh Pappu, Haotian Ye, Wanjia Zhao, Surya Ganguli, and James Zou. Physics of Agents: Statistical mechanics predicts collective behavior of AI agents. arXiv:2608.16578, 2026a.
- Batu El, Jinhee Paeng, Fatih Dinc, Shiye Su, Mete Erdogan, Aneesh Pappu, Haotian Ye, Wanjia Zhao, Surya Ganguli, and James Zou. Physics of Agents: data and code [dataset]. GitHub repository, <https://github.com/batu-el/physics-of-agents>, 2026b. Commit f571742, accessed 2026-08-18; Apache License 2.0.
- Ariel Flint, Luca Maria Aiello, Romualdo Pastor-Satorras, and Andrea Baronchelli. Group size effects and collective misalignment in LLM multi-agent systems. *Proceedings of the National Academy of Sciences*, 123(34):e2531697123, 2026. doi: 10.1073/pnas.2531697123.

- Ariel Flint Ashery, Luca Maria Aiello, and Andrea Baronchelli. Emergent social conventions and collective bias in LLM populations. *Science Advances*, 11(20):eadu9368, 2025. doi: 10.1126/sciadv.adu9368.
- Makoto Fukushima. Analytically tractable model of synaptic crowding explains emergent small-world structure and network dynamics. *Scientific Reports*, 16:11748, 2026. doi: 10.1038/s41598-026-47213-2.
- Nathan O. Hodas and Kristina Lerman. How visibility and divided attention constrain social contagion. In *Proc. 2012 ASE/IEEE International Conference on Social Computing (SocialCom)*, 2012.
- Yibo Hu and Jiaming Qu. Social pressure breaks majority voting in LLM safety panels. arXiv:2608.04415, 2026.
- Mrinank Sharma, Meg Tong, Tomasz Korbak, David Duvenaud, Amanda Askell, Samuel R. Bowman, et al. Towards understanding sycophancy in language models. arXiv:2310.13548, 2023.
- Tadahiro Taniguchi, Shiro Takagi, Jun Otsuka, Yusuke Hayashi, and Hiro Taiyo Hamada. Collective predictive coding as model of science: Formalizing scientific activities towards generative science. *Royal Society Open Science*, 12:241678, 2025. arXiv:2409.00102.